

YIZHOU DING

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EDUCATION

Nanyang Technological University

MSc in Financial Engineering

Singapore

Fall 2026

Southwestern University of Finance and Economics (SWUFE) & University of Delaware (UD)

Dual Bachelor's Degree in Financial Mathematics & Finance; GPA: 90.75/100

Chengdu, China

2022 – 2026

- **Relevant coursework:** Multivariate Calculus, Linear Algebra, Probability Theory, Ordinary Differential Equations, Mathematical Statistics, Econometrics, Optimization, Stochastic Processes, Machine Learning, Financial Stochastic Analysis, Data Structures.

MANUSCRIPTS UNDER REVIEW

- Ding, Y., Zhou, X., & Lyu, P. “A Variable-Step Crank-Nicolson and High-Order Compact Scheme for a Nonlinear Black-Scholes Type Equation.” Manuscript under review.

RESEARCH EXPERIENCE

A Variable-Step Crank-Nicolson and High-Order Compact Scheme for a Nonlinear Black-Scholes Type Equation

Chengdu, China

Research Assistant, Supervisor: Prof. Pin Lyu, SWUFE

Sep. 2024 – Present

- Conducted a structured literature review on compact finite-difference schemes, variable-step Crank-Nicolson methods, and nonlinear Black-Scholes equations with counterparty risk.
- Assisted in formulating a fully discrete linear scheme that combines variable-step temporal approximation, fourth-order compact spatial discretization, and explicit approximation of nonlinear terms.
- Worked through key components of the stability and convergence analysis using discrete energy estimates, contributing to the derivation of the optimal error bound of $\mathcal{O}(\tau^2 + h^4)$.
- Implemented MATLAB experiments to examine convergence rates, stability performance, and computational efficiency under alternative mesh and time-step settings.
- Contributed to manuscript preparation, including literature synthesis, numerical result presentation, and revision of the theoretical exposition; manuscript currently under review.

Database Design & Implementation

Chengdu, China

Teaching Assistant, Supervisor: Prof. Leting Zhang, UD

Feb. 2026 – Jun. 2026

- Assisted course delivery on relational database design, SQL implementation, and data-management concepts, supporting students in translating database theory into executable queries and project work.
- Provided guidance on assignments and course projects, helped troubleshoot implementation issues, and supported grading and feedback preparation.

People's Bank of China: Global Central Bank Digital Currency Study

Remote / China

Research Contributor

Jun. 2023 – Dec. 2023

- Reviewed global CBDC initiatives and the Digital Euro investigation phase, synthesizing technology design, adoption channels, policy motivations, and regulatory implications.
- Connected technical design choices with broader questions in digital finance, institutional trust, payment infrastructure, and future financial intermediation.

Automatic Event Scheduling Based on LLMs

Chengdu, China

Leader of Team Project, Instructor: Prof. Leting Zhang, UD

Feb. 2025 – Apr. 2025

- Built a prototype scheduling system integrating an SQL-based venue database with LLM interpretation of natural-language constraints for multi-constraint event planning.
- Localized DeepSeek models using Ollama and developed a Flask web interface, strengthening preparation for research at the intersection of AI, digital platforms, and business applications.
- Selected by the course instructor as an exemplary project to support the learning of future cohorts.

INDUSTRY AND APPLIED RESEARCH EXPERIENCE

SWS Futures

Shanghai, China

Quantitative Research Intern

Jul. 2025 – Sep. 2025

- Extended the Nelson-Siegel framework by incorporating multidimensional macroeconomic factors and XGBoost, investigating whether macro information could explain yield-curve movements not captured by conventional latent factors.
- Refined a moving-average RSI signal through standardization, hyperparameter tuning, and systematic backtesting diagnostics, assessing its robustness and transferability across agricultural futures rather than relying on performance from a single contract.
- Synthesized monetary-policy scenarios, supply-demand conditions, and sector fundamentals in the black-metals market, translating macroeconomic shocks into exposure analysis and hedging implications for steel companies.

Galaxy Futures

Shanghai, China

Futures Analyst Intern

Dec. 2024 – May 2025

- Formulated and tested hypotheses on intraday volatility patterns using high-frequency crude-oil and soybean-meal options data, separating empirical pattern identification from strategy evaluation under explicit backtesting assumptions.
- Developed a localized LLM-assisted forecasting pipeline that integrated price-volume indicators, sentiment information, and fundamental and macroeconomic variables, gaining experience in organizing heterogeneous data into a coherent modelling workflow.
- Compared minimum-variance futures hedging, one-to-one futures hedging, and options-based hedging within a consistent analytical framework, examining their trade-offs in hedging effectiveness, cost, and practical implementation for rebar exposure.

Futurus Vessel Capital

Shanghai, China

Risk Analyst Intern

Jul. 2024 – Aug. 2024

- Implemented reusable Python routines for VaR, drawdown, annualized return, and scenario-based stress testing, building a quantitative foundation for evaluating portfolio risk from multiple perspectives.
- Applied moving averages, ARIMA models, turnover-based volatility proxies, and simulation methods to financial time series, testing whether observed relationships remained consistent across sectors, market-cap groups, and investment styles.
- Built a dashboard linking fund managers' trading activity with contemporaneous market movements, translating quantitative outputs into an interpretable tool for examining investment preferences and risk exposure.

Industrial and Commercial Bank of China

Inclusive Finance Intern

Yangzhou, China

Jul. 2023 – Aug. 2023

- Examined regional economic indicators and rural financial-service coverage to understand how local economic conditions, institutional capacity, and product availability affected access to financial services.
- Translated the analysis into practical recommendations on service coverage, product design, and institutional support, developing an early understanding of how financial decisions must account for implementation constraints and heterogeneous local needs.

LEADERSHIP & COMMUNITY SERVICE

University Broadcasting Station

Station Manager

Chengdu, China

Sep. 2022 – Jun. 2025

- Led programming, host coordination, and editorial planning for campus broadcasting, balancing content quality, team training, and operational continuity.
- Organized student communication projects and cross-department collaboration, strengthening public-speaking, event-planning, and leadership experience.

SWUFE Study Abroad Club

Student Chairman

Chengdu, China

Sep. 2022 – Jun. 2026

- Organized academic seminars and networking events with scholars including Prof. Leila Pinto Campillo (UCL), Prof. Patrick Rebeschini (Oxford), and Prof. Miquel Noguer Alonso (NYU).
- Led workshops on graduate-school applications and academic planning, supporting knowledge exchange among students interested in overseas study.

Tri-Service to the Countryside Program

Publicity Team Member, “Data Empowerment and Technology-Enabled Agriculture” Project

Jinchuan County, Sichuan, China

Aug. 2023 – Nov. 2023

- Conducted field visits, educational outreach, and digital-commerce workshops in rural communities, helping local residents understand e-commerce, logistics management, and data-driven agricultural sales.
- Led the writing and documentation of the team’s community-service activities; the resulting field report was published by *China Youth Network* and featured by local government media.
- Contributed to a project that received the University First Prize in the Tri-Service to the Countryside social-practice program.

AWARDS & SCHOLARSHIPS

Awards

- “Xuechuang” Cup, National Second Prize
- 10th Arts Exhibition of Sichuan Province, Provincial First Prize
- Tri-Service to the Countryside, University First Prize

Nov. 2024

Aug. 2023

Nov. 2023

Scholarships

- Merit Student
- Outstanding Student Worker
- Academic Scholarship (Top 10%)

2022 – 2026

2022 – 2026

2022 – 2026

SKILLS & INTERESTS

- **Languages:** Mandarin Chinese (native); English (IELTS 7.5; GRE 326).
- **Programming Languages:** Python, C++, LaTeX, MATLAB, R, SQL, HTML.
- **Software and Tools:** Codex, Zotero, VS Code, EViews, Flask, Ollama, Microsoft Office, Git.
- **Interests:** stage acting, table tennis, swimming.